
Auditable Compositional Question Answering over UK Public Procurement

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Abstract

UK public procurement records are public and partly structured, yet reliable analysis remains difficult. Organisation identifiers are fragmented, monetary fields carry different accounting meanings, and useful questions require several joins, filters, and aggregations. We build an auditable question answering framework over 166,277 Open Contracting Data Standard releases. Questions become programs in a closed typed algebra, executed over a normalised data layer under deterministic conventions, with abstention when a question is ambiguous, unsupported, or matched by no records. The resulting benchmark contains 12,828 questions with executable gold programs, and a second evaluator, written without shared execution code, reproduces 99.88% of 14,770 audited oracle cases.

Experiments separate the value of the executable scaffold from the value of model training. On development data, an untrained 8B model rises from a 31.5% retrieval baseline to 70.4% when routed through the scaffold. Training raises the strongest fully local system to 85.65% on a sealed 2,285-question test, above the 69.76% cloud teacher that proposed its training candidates, and the ordering holds with a second base model and on a sealed challenge set, 78.31% against 50.33%. Sealed access baselines locate the mechanism, retrieval answers single-record questions yet collapses on exhaustive aggregation even when the gold plan is supplied as the query. A transfer to WikiTableQuestions raises an untrained 22.51% baseline to 44.43% with answer-only supervision and 51.80% with gold programs, and the transfer audit identifies representation and program-language coverage as the main remaining bottleneck.

1 Introduction

Large language models can interpret natural-language questions and produce fluent answers, but they remain unreliable when a task requires several linked operations. A model may retrieve the right records while applying the wrong relation, counting the wrong unit, or aggregating values that should remain separate. Structured query systems offer a different strength. Their entities, relations, and operations can be inspected and executed again. Their weakness is coverage, since they depend on a defined schema and a restricted query language. Reliable question answering over structured data therefore requires both flexible language understanding and a reproducible path from evidence to answer.

UK public procurement is a demanding setting for this problem. Contract notices contain buyers, suppliers, awards, dates, locations, and monetary values, which makes them suitable for factual, temporal, and compositional analysis. The records are not clean enough to query directly. One organisation can appear under many source identifiers and name variants. Values may be reported at tender, lot, award, and contract level, where some fields represent separate amounts and others repeat the same amount. A program can therefore be valid and executable while still computing the wrong quantity. Correctness depends on the program, the data conventions, and the link between the two.

Prior work addresses parts of this setting. Retrieval-augmented generation can supply relevant records, but it does not guarantee correct joins, counts, or aggregations. Neural-symbolic question answering maps language to executable programs, but is usually evaluated over curated schemas and stable execution environments

(Cheng et al., 2023; Gu et al., 2023; Luo et al., 2024a). Methods such as STaR and rejection-sampling fine-tuning learn from selected model outputs (Zelikman et al., 2022; Yuan et al., 2023; Gulcehre et al., 2023). These methods show that executable or answer-based feedback can support training. They do not resolve the full problem considered here, where the source semantics must first be audited, refusal is part of the task, and successful execution does not by itself establish that the program answers the question.

We build the task and system end to end. Five years of procurement releases are converted into a canonical data layer with explicit rules for entity identity, value provenance, and monetary additivity. Questions are compiled into a small typed algebra. A language model proposes the interpretation and program. Deterministic code then checks the program, grounds it to the data, and executes it. During training, sampled programs are retained only when their executed answers match the benchmark oracle. During deployment, no oracle is available or required. The final answer comes from execution, not free-form model generation.

The evaluation asks three questions. First, how much of the performance comes from the executable scaffold, and how much comes from task-specific model training? Second, does the resulting capability survive new program structures and question surfaces? Third, when the architecture is moved to another structured-data domain, which component limits performance? The experiments show that the scaffold establishes a strong floor, training both question understanding and planning improves generalisation, and cross-domain transfer is limited mainly by representation and program-language coverage.

Contributions. This work makes three contributions.

1. We construct an auditable procurement question-answering task with explicit data conventions, executable gold programs, scored abstention, and answer oracles checked by an independently implemented evaluator.
2. We develop a typed executable framework that separates language understanding from deterministic computation. The same framework supports program training from teacher and self-generated candidates without making the language model the source of the final answer.
3. We provide a controlled empirical decomposition across retrieval, executable scaffolding, local planning, sealed external evaluation, and table-domain transfer. The analysis identifies where performance comes from and where the closed language still limits coverage.

2 Related Work

Executable question answering. Semantic parsing and knowledge-base question answering translate natural language into logical forms that can be executed against structured knowledge. Recent systems improve coverage through in-context program generation, graph-constrained search, retrieval-based grounding, or tool use (Cheng et al., 2023; Gu et al., 2023; Luo et al., 2024a;b; Jiang et al., 2023). Table question-answering systems use SQL, Python, neural table encoders, or hybrid execution (Pasupat & Liang, 2015; Herzig et al., 2020; Yin et al., 2020; Liu et al., 2022; Jiang et al., 2022; Ye et al., 2023; Ni et al., 2023; Wang et al., 2024). These methods establish the value of executable intermediate representations. Our setting adds a problem that curated benchmarks usually remove. The source fields and entity identities are themselves uncertain, so execution is only meaningful after the data conventions have been specified and audited.

Learning from generated programs. STaR, rejection-sampling fine-tuning, and ReST generate candidate solutions, retain those that satisfy an acceptance rule, and train on the survivors (Zelikman et al., 2022; Yuan et al., 2023; Gulcehre et al., 2023). V-STaR learns a verifier from correct and incorrect solutions, while executable SQL systems use answer or execution feedback to build training data (Hosseini et al., 2024; Zhang et al., 2025). Weak-to-strong work studies when a student can generalise beyond imperfect supervision (Burns et al., 2024). We do not claim the generate-filter-train loop as a new method. We use it as one part of an executable system and study how it behaves when a program can compile and run while expressing the wrong analysis.

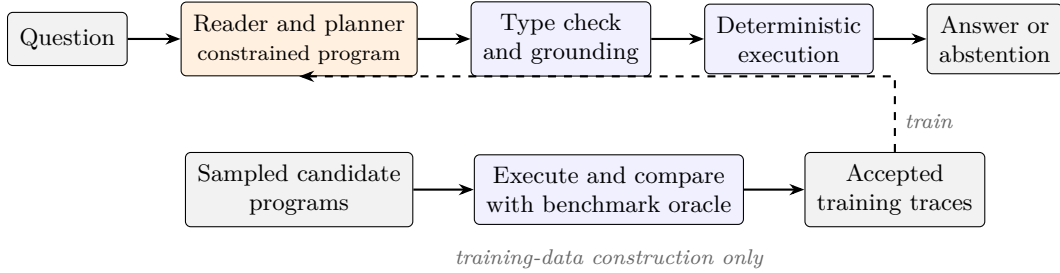


Figure 1: System overview. The deployment path ends with deterministic execution. Oracle comparison is used only when constructing training data.

Procurement data, provenance, and refusal. Procurement knowledge graphs such as TheyBuyForYou support data integration, search, and risk analysis across public records (Soylu et al., 2022). Our focus is a measurable natural-language task rather than a graph resource alone. Each answer is tied to an executable program and grounded records. Questions that are ambiguous, unsupported by the data model, or matched by no records are represented explicitly, in the spirit of unanswerable-question benchmarks and selective prediction (Rajpurkar et al., 2018; Kamath et al., 2020), but scored against executable programs rather than span annotations. This connects provenance to the execution trace and makes failure detection part of the benchmark rather than an informal property of the interface.

3 Task and Method

3.1 Overview

Figure 1 separates deployment from data construction. At deployment time, the model reads a question and emits a program under constrained decoding, and deterministic components check, ground, and execute it, returning an answer or abstention. During training-data collection, candidates sampled at positive temperature enter the training pool only when their executed answers match the benchmark oracle, so oracle access is a property of the data pipeline, not the runtime system.

3.2 Data conventions and canonical representation

The source corpus contains 166,277 compiled releases in the Open Contracting Data Standard (Open Contracting Partnership, 2024), from 2022 to 2026. A field audit identified two issues that determine many downstream answers. First, signed values are often stored in contract sub-records rather than the nominal award-value field. Second, most buyer and supplier references are platform account identifiers. One real organisation can therefore appear under many IDs. The Ministry of Defence appears under 77 platform identifiers in one year of the source data.

We convert the releases into a canonical representation with 215,221 award records and 131,502 organisations. Entity resolution is precision-first. Registry identifiers and safe deterministic variants are merged before normalised name and regional evidence are considered. Fuzzy matching produces review candidates but never writes a merge directly. Monetary values follow a fixed fallback rule and retain their source field and an additivity flag. The flag prevents repeated framework ceilings from being summed once for every linked award. Logically, the representation is a typed property graph whose nodes represent procurement entities, contracts and classifications, and whose edges represent contractual and organisational relations. For execution it is materialised as typed columnar tables rather than stored in an RDF or property-graph database, since the columnar form supports vectorised execution, unit tests, and a second evaluator that shares no execution code with the main system.

3.3 Program language and deterministic execution

Questions are represented as trees in a closed typed algebra. Seven value types and seventeen procurement operators cover filtering, projection, counting, guarded summation, grouping, ranking, scalar comparison, and set operations. Each node has an explicit input and output type. The complete inventory appears in Appendix A. One sealed PACS question illustrates the full path from language to answer.

Compare two slices, works notices during 2025 on one side, services notices during 2022 on the other. Does the first side hold more notices?

```
combine[gt](count(filter(category=works, year=2025)),  
            count(filter(category=services, year=2022)))
```

The program type-checks to `BOOL`, both branches execute over the full record universe, and the executed answer is `False`, which matches the oracle. The answer arrives with the record sets of both branches, so a reviewer disputes the program, not a paragraph.

The algebra is narrower than SQL or Python, but it provides three useful properties. Membership is decidable by a recursive type checker. The grammar can be compiled into a JSON schema and enforced through structured decoding in the serving layer’s guided-generation interface (Dong et al., 2024). The operator set is small enough for a second evaluator to implement independently. After a program is produced, the grounding layer resolves entities, fields, dates, and values to the canonical representation. The executor applies contract-level deduplication, exact decimal arithmetic, the additivity guard, empty-key exclusion, and deterministic tie breaking. It returns both the answer and the records used to compute it.

3.4 Planning, training, and abstention

The procurement system uses two learned stages. A reader extracts the requested answer type, question literals, and dependency structure. A planner compiles the question and brief into a program. This split was retained because it outperformed direct generation on procurement, and the margin is quantified in Section 5.1. The WikiTableQuestions transfer uses a fused reader-planner because fixed handoffs were weaker during table development.

Training candidates are sampled from a cloud teacher or from the local student. A candidate is retained when it has a valid type structure, grounds to the question and data, executes successfully, and produces the oracle answer. The oracle does not author the candidate and is not included in the model input. The training ladder includes supervised fine-tuning, self-harvest with rejection sampling, and preference optimisation, the last excluded from the primary systems because its near-miss negatives were unstable across the two bases.

Abstention is part of the output space. The system can return ambiguous, unsupported, or no-result when the question or evidence does not justify an answer. A repair step is available only after a typed failure. Refusals and meaningful empty results are final, since an earlier ungated repair design changed six correct refusals into incorrect answers.

3.5 Benchmark construction and oracle audit

The main benchmark is generated program first. Parameterised programs are instantiated and executed before a question surface is written, so every row is created with an executable gold program and answer. The benchmark contains 12,828 questions, including answerable, ambiguous, unsupported, and empty-result cases, split into 9,267 training rows, three development slices, and a sealed test of 2,285 rows. Splits separate program structures rather than surface wording. Mechanical gates prevent plan overlap, duplicate identifiers, local sampling omissions, and multi-answer ambiguity.

A second evaluator reconstructs answers from the raw releases without shared execution code. Across 14,770 audited oracle cases, the two implementations agree in 99.88% of cases. The audited set comprises every answerable oracle produced by the benchmark generator at audit time, and the final curated benchmark of 12,828 rows, which also contains refusal rows carrying no value oracle, draws from this audited pool. All

18 remaining disagreements were inspected, and none affects a reported comparison. The audit measures implementation agreement under the declared conventions rather than removing uncertainty from the source records themselves.

4 Evaluation

Benchmarks. The main evaluation uses a sealed test of 2,285 procurement questions. PACS contains 922 rows from seven procurement-analysis families and tests whether the system depends on the surface generator used for the main benchmark. Its specification, question grammar, and logical-shape labels were frozen before the model was evaluated. Five isolation checks cover text, program hash, template identity, entity anchors, and logical shape, each sealed intent is rendered through an independently written surface grammar, and naturalised rewrites pass deterministic logical-signature checks against the gold program. WikiTableQuestions contains 22,033 crowd-written questions over 2,108 web tables (Pasupat & Liang, 2015). Development uses table-disjoint folds, and the official 4,344-row test is evaluated once under a recorded manifest.

Systems. Five main systems differ in where question reading and program planning run, over a shared data layer and executor. The cloud teacher reads and plans through its provider API zero-shot. The two hybrid systems read with the cloud model and plan with a trained local model, and the two fully local systems use trained local models for both stages. The local models are Qwen3-8B and Llama-3.1-8B, trained with rank-64 LoRA under 4-bit QLoRA (Hu et al., 2022; Dettmers et al., 2023). All local systems, including the untrained baselines, use the same executor and constrained decoder, which prevents output-format errors from dominating the comparisons. The cloud teacher uses its provider API and cannot use the same token-level grammar constraint, so teacher comparisons are reported as deployable-system comparisons rather than equal-decoding model comparisons. Three further sealed baselines vary data access while holding the answering model fixed at the teacher class. A closed-book run answers from parametric knowledge alone, a retrieval reader answers from the top retrieved records, and a gold-constraint variant rebuilds the retrieval query from the gold program’s constraint literals. Each is a frozen configuration evaluated in a single sealed pass.

Protocol. Each primary evaluation uses one temperature-zero model call and no repair. Answerable questions use type-aware exact match against the audited oracle. Refusal questions use strict status match and a faithfulness-gated metric that checks whether every literal in the emitted program is supported by the question. Paired systems are compared with exact McNemar tests (McNemar, 1947). PACS intervals use cluster bootstrap over intent clusters. The cloud teacher cannot be seeded, so its variability was measured with three identical development runs at 72.6 ± 1.1 points, and no teacher comparison below three points rests on a single run. Each student is a single frozen training run. Paired significance therefore measures test-set disagreement and not training-seed variance. Full training configurations, serving settings, compute footprint and the audit rule behind every reported rate appear in Appendix D.

5 Results

5.1 Where the procurement performance comes from

This subsection makes two moves, a decomposition on development data, since intermediate arms never touch the sealed test, and a sealed confirmation of the frozen champion systems. Figure 2 shows both, and Table 6 in the appendix lists every rung.

On development, the retrieval baseline reaches 31.5%, and an untrained Qwen3-8B routed through the typed program interface and executor reaches 70.4%, already close to the cloud teacher’s replicate band. This first step shows that exhaustive computation and a fixed execution contract provide much of the initial gain before any training. Three training arms then branch from the supervised model rather than forming a sequence. Rejection-sampling fine-tuning on the student’s own verified survivors, with the teacher absent, holds level with supervised fine-tuning, so the verified pool rather than continued teacher access carries the signal. Preference optimisation splits by base, 83.5% on Qwen against 77.3% on Llama, the instability that

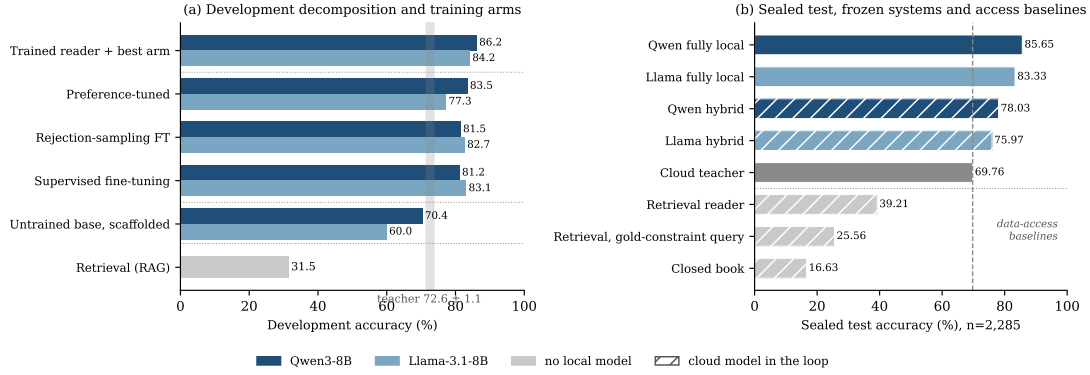


Figure 2: Procurement results. Colours denote the base model; hatching in panel (b) marks cloud-reader systems. (a) Development decomposition across the executable scaffold and the training arms, with the shaded band covering three teacher runs. (b) Sealed-test results. The five frozen systems sit above the three sealed data-access baselines, each a single sealed pass, with the answerable and refusal split in Table 5 of the appendix. The untrained scaffolded base is measured sealed on the challenge set in Table 1.

removed it from the primary systems. The fully local rung changes a different axis, replacing the cloud reader with a trained local one on top of the best arm per base, and reaches 86.2% and 84.2% on the two bases.

A 100-question format diagnostic, reported in Appendix B, confirms that constrained decoding equalises output format across systems, so accuracy differences under the shared decoder measure planning rather than output discipline.

In the sealed confirmation of Figure 2(b), the strongest local system is 15.89 points above the teacher, with a 95% paired interval of 14.07 to 17.70 and exact McNemar $p < 10^{-14}$. The same ordering holds with the second base, which reduces the chance that the result is specific to one model family. Fully local systems also outperform the hybrids on both bases, by 7.61 points with interval 6.10 to 9.13 on the stronger base, and the cross-base version of the comparison is sharper still. The fully local system on the weaker base outperforms the hybrid on the stronger base by 5.30 points with interval 3.62 to 6.97, so trained question reading contributes more than base capability on this task.

Figure 5 in the appendix resolves the sealed result by question type across all systems, which turns the aggregate ordering into a mechanism. The teacher’s weakness concentrates exactly where exhaustive computation lives, 48.7% on counts and 61.3% on sums against 76.2% and 92.6% for the strongest local system, which is the failure retrieval-style reasoning cannot escape. Comparison questions expose a base-dependent skill, with the two Qwen systems at 82.1% and 85.8% against 40.6% and 44.3% for the Llama systems. And the refusal columns close the abstention claim on the main benchmark, where the strongest local system recognises ambiguous, empty-result and unsupported questions at 99.2%, 100.0% and 100.0%.

Data access ablation. The three access baselines in Figure 2(b) isolate what the executable evidence path contributes beyond the answering model, and Table 5 in the appendix gives the finer split. Answerable accuracy is separated from refusal recognition because the weak baselines earn refusal credit degenerately, a model that cannot find an answer replies unknown, and unknown scores as a refusal. The closed-book run pairs a perfect refusal column with 1.0% on answerable questions, so the corpus is not recoverable from parametric knowledge. The retrieval reader reaches 66.5% on single-record questions yet collapses to 3.7% on the six aggregation families. The gold-constraint variant removes planning as a failure source and aggregation stays at 3.3%, so the failure is the enumeration limit of top- k evidence rather than query formulation, and it is reported for attribution only since its injected literals disturb lexical matching elsewhere. The fully local system inverts the pattern, scoring higher on aggregation than on single-record questions, because exhaustive execution turns the families that defeat retrieval into the most reliable ones.

Table 1: Sealed PACS results by task family and status class, computed for all systems on identical row sets. Family rows use the 655 answerable rows with concrete value oracles. The refusal row uses the 138 unsupported-question golds. The empty-result class was affected by the documented 46-row labelling defect, and only the planner was re-scored under the corrected labels, marked †. The retrieval baseline was not run on status rows. Bold marks the best system per row.

	n	Local planner	Cloud teacher	Untrained base	Retrieval
F1 spending and volume	104	99.0	69.2	59.6	12.5
F2 temporal change	107	96.3	81.3	60.7	53.3
F3 supplier concentration	97	47.4	38.1	26.8	11.3
F4 cross-buyer comparison	107	93.5	43.0	54.2	34.6
F5 overlap and exclusion	103	97.1	21.4	20.4	3.9
F6 relational composition	88	88.6	31.8	36.4	18.2
F7 disclosure and compliance	49	38.8	40.8	34.7	4.1
All answerable, value oracles	655	83.8	47.6	42.9	21.4
Refusal, exact status	138	100.0	91.3	29.7	–
Empty result, corrected labels	90	38.9 [†]	–	–	–

5.2 Distribution shift and a generator-independent procurement benchmark

Comparing the two panels of Figure 2 system by system gives the distribution-shift result directly. The hybrid systems lose 5.5 and 7.1 points from development to test, while the fully local systems lose 0.5 and 0.9, and the pattern holds on both bases. The decay asymmetry is consistent with the fully local system depending less on the cloud reader and the development surface distribution. It does not by itself establish a causal robustness mechanism.

PACS tests new question surfaces and logical exposure. The frozen local planner reaches 78.31% strict accuracy on 922 sealed rows and 80.00% on the answerable subset, against 50.33% for the cloud teacher, 36.01% for the untrained base, and 21.90% for retrieval under the same strict metric, so every rung of the deployment ladder is present on questions none of the systems authored. The teacher API lacks the local constrained decoder, and 23.1% of its rows fail on format, so the margin is not presented as a decoding-neutral model comparison.

Table 1 breaks the sealed result into task families and status classes on identical aligned row sets, whose denominators differ from the strict overall by construction. The local planner leads six of the seven families, most extremely on overlap and exclusion, 97.1% against 21.4% for the teacher, whose free-form programs rarely survive the set-operation semantics. The single exception is disclosure and compliance, the weakest family for every system, where the teacher edges the planner 40.8% to 38.8%. Universal quantification over every notice of a buyer defeats all systems, and the current algebra and training data do not resolve this task family. On the refusal axis the planner recognises every one of the 138 unsupported sealed questions, against 91.3% for the teacher and 29.7% for the untrained base, so scored abstention is not only a design feature but a measured sealed-test behaviour.

The PACS audits, shown in Figure 3 of the appendix, separate depth from novelty. A row is irreducible when deleting any predicate from its gold program changes the answer. This holds for 42.7% of answerable rows, and the planner scores 78.7% on that subset, close to the 80.0% answerable result. Intent shapes not shown during training reduce accuracy to 68.5%, a cost of 11.5 points with a 95% cluster bootstrap interval of 5.6 to 17.2. The system therefore transfers new compositions of familiar constructs more reliably than entirely new logical forms, and Appendix B locates the weak cells precisely.

5.3 Transfer to WikiTableQuestions

The transfer retains the typed execution architecture and replaces the procurement data adapter with table views, a column-aware value linker, and one row-preserving arg-extremum operator. Table 2 reports the official test results. The procurement checkpoint carries a 4.82-point zero-shot gain over the untrained base.

Table 2: WikiTableQuestions results. Official sealed test (n=4,344, one pass under a recorded manifest), each rung significant over the previous at exact McNemar $p < 5 \times 10^{-18}$, alongside the development fold before and after representation and compiler closure.

	Official test	Step gain	Dev fold	after closure
Untrained local base	22.51			
Procurement checkpoint, zero-shot	27.33	+4.82		
Answer-only training	44.43	+17.10	31.7	40.0
Gold-program training	51.80	+7.37	40.0	51.0

Answer-only training reaches 44.43%, and gold-program training reaches 51.80%. The answer-only result is comparable with classical weakly supervised parsers, but both trained systems remain below recent specialised table-QA methods (Cheng et al., 2023; Ye et al., 2023; Wang et al., 2024). The experiment is used to study portability and bottlenecks, not to claim state of the art.

A differential audit against 11,276 SQUALL SQL programs (Shi et al., 2020) locates the remaining gap. The original algebra expresses 53.9% of the reference programs, compiler closure and typed loader views raise coverage to 61.0%, and the remaining 39.6% lie outside the language. Figure 4 in the appendix attributes the recovered programs to each mechanism. Translated in-coverage programs match the reference denotation in 92–93% of cases, and execution recovers the benchmark answer in 94.1% of those cases. The reference programs are themselves imperfect, reproducing the benchmark answer in 88.9% of cases, so the audit treats them as a noisy reference rather than an absolute standard. This pattern points to coverage rather than executor correctness as the main transfer limit.

The coverage account is also tested by intervention. After closure and retraining, both supervision regimes improve, by 8.3 and 11.0 development points in Table 2, so the transfer gap is not solely a planning-model problem.

6 Discussion and Limitations

The results support three conclusions. A deterministic execution contract changes the problem substantially, since an untrained local model becomes competitive with the cloud teacher once both operate through the typed interface. Task-specific training still matters, since the fully local systems outperform their hybrid counterparts and degrade less from development to test. And portability is bounded by what the adapter, compiler, and algebra can represent, the boundary the WikiTableQuestions audit measures directly.

The scale of the evidence supports system-level conclusions beyond a small prototype, a full five-year national corpus, an independently verified benchmark, and one pass on each of two sealed test sets. The resulting system could support a local procurement-analysis workflow on a single GPU, answering exhaustive aggregation questions that retrieval cannot, refusing when the data cannot support an answer, and returning programs a reviewer can execute rather than prose a reviewer must trust.

The teacher comparison is task-specific, an in-domain student against a zero-shot deployment, so it shows that the full local system is more accurate on the defined task, not that an 8B model has greater general capability. PACS strengthens the external validity claim, but it is still a controlled benchmark rather than a sample of real analyst questions.

Five limitations remain. The main benchmark is program-first and its question surfaces are generated under style controls. Each student result comes from one training run, so training-seed variance is not estimated. The teacher and local systems do not share identical decoding constraints. The closed algebra rejects questions outside its supported operations, and unseen intent shapes remain a clear weakness. The transfer study lacks a baseline in which a language model writes SQL that is executed directly, a comparison that belongs to future work on development folds now that the official test is consumed. Finally, the study does not simulate arbitrary deletion of KG facts, so the refusal tests cover missing or unsupported evidence but not every form of data incompleteness.

Refusal did not transfer free of charge. On a paired comparison against the earlier production system, the compositional planner wins the answerable subset by 4.4 points yet loses explicit abstention clearly, because its training mix carried only 37 refusal demonstrations, so under the pre-registered bar of no key-bucket regression, no retirement claim is made for the older system. Competence follows demonstrations, and refusal is a competence like any other.

The candidate-selection experiment in Appendix C adds a negative result. An LLM selector does not improve over the first accepted candidate, while a small deterministic scorer on execution features gains 2.0 points, a finding that concerns best-of- k selection only.

7 Conclusion

We presented an auditable question-answering framework for UK public procurement, combining explicit data conventions, typed program generation, deterministic execution, abstention, and independent oracle audit. The experiments show that the executable scaffold establishes a strong base, training both reading and planning improves generalisation, the result holds on sealed and generator-independent tests, and representation and program-language coverage are the main remaining limits.

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A Operator Algebra

Programs are trees over seven value types. RECORDS is a subset of the record universe, VALUES is a distinct set of scalar values, GROUPS maps group keys to numbers, and RANKING is an ordered list of key-value pairs. NUMBER, VALUE, and BOOL are scalar types. Program depth is capped at 16 and node count at 64. Each checker failure contains a typed diagnostic and a path into the program tree.

Table 3: Closed operator inventory. The WikiTableQuestions transfer adds the row-preserving **extreme_rows** node.

Node	Signature	Function
filter	predicates $\rightarrow$ RECORDS	filter records
values	RECORDS $\rightarrow$ VALUES	project distinct values
count	RECORDS $\rightarrow$ NUMBER	count records
size	VALUES $\rightarrow$ NUMBER	set cardinality
sum	RECORDS $\rightarrow$ NUMBER	guarded money sum
exists	RECORDS $\rightarrow$ BOOL	test non-emptiness
select	RECORDS $\rightarrow$ VALUE	unique field lookup
extreme	RECORDS $\rightarrow$ VALUE	record argmin or argmax
groupby	RECORDS $\rightarrow$ GROUPS	grouped count or sum
argext	GROUPS $\rightarrow$ VALUE	extremum group key
top	GROUPS $\rightarrow$ RANKING	top- k groups
num	literal $\rightarrow$ NUMBER	numeric literal
combine	NUMBER ² $\rightarrow$ scalar	arithmetic or comparison
vcompare	VALUE $\rightarrow$ BOOL	scalar comparison
setop	VALUES ² $\rightarrow$ VALUES	set operation
gcombine	GROUPS ² $\rightarrow$ GROUPS	combine grouped values
keys_where	GROUPS $\rightarrow$ VALUES	thresholded group keys
extreme_rows	RECORDS $\rightarrow$ RECORDS	row-preserving arg-extremum

B Audit and Supplementary Results

Entity resolution. The source contains 204,711 raw organisation aliases. Resolution proceeds through registry identifiers, safe deterministic variants, normalised name and region, and finally name-only evidence. A total of 255 borderline cases were placed in a review file, and 82,013 singletons were left unmerged. Similarity matching never writes a merge directly.

Benchmark gates. Every generated program is re-executed against the full graph before its question is admitted. The build rejects local sampling omissions, multi-answer ambiguity, plan overlap, duplicate identifiers, and row-count changes. A position-aware mechanical detector checks whether paraphrases preserve the logical signature. Nineteen of the 2,285 final-test rows contain phrases also seen in refusal traps. Removing them changes every headline by at most 0.21 points and changes no ranking.

PACS isolation. Five zero-overlap checks hold between training and sealed evaluation at the level of question text, logical signature, template identity, entity anchors, and program hash. A 93-row human audit took place before unsealing. It found one systematic boolean error affecting 46 rows, which was corrected by offline re-execution before any model score was read.

Format diagnostic. Decoded without constraints on 100 development questions, the untrained base produces question-grounded plans with real entities in an invented format on 98 and matches the executable contract on 3, while the tuned models match the contract on 98 and 99. Fine-tuning therefore teaches conformance to the specific contract, and constrained decoding grants that contract to every system.

PACS weak cells. Beyond the family rows of Table 1, the deep cells of supplier concentration, which bind a scope inside each group, fall to between 15% and 30%, while the five families outside disclosure and concentration score between 82% and 97% on value-oracle rows. The repair arm in Figure 3 is a development-only variant and is not part of the sealed headline.

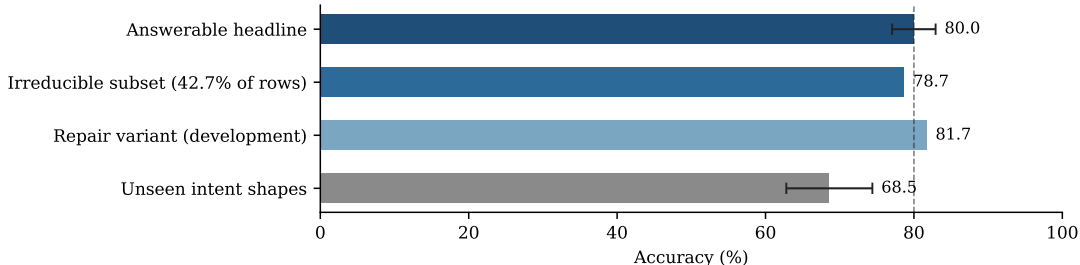


Figure 3: PACS audits. The irreducible subset remains close to the answerable headline, while unseen intent shapes produce the largest measured loss. The repair arm is a development-only variant.

PACS composition and status axis. Table 4 gives the exact partition of the sealed set. The 267 rows whose serialised oracle is empty are the 138 unsupported golds, the 90 empty-result golds, and 39 answerable rows whose oracle serialisation failed at build time. The planner recognises all 138 unsupported golds, is credited 35 of the 90 empty-result golds after the documented relabelling, and scores zero on the 39 unserialised rows, which reconciles exactly with the reported overall, 549 answerable plus 138 plus 35 equals 722 of 922. Per-family and status-class results appear in Table 1 of the main text.

Training ladder. Table 6 reports development accuracy and data-engine acceptance rates. Acceptance rate is the fraction of sampled answerable candidates that pass the data pipeline, over the 8,555 answerable training questions. The teacher pass accepts 5,605 traces, and the first self-harvest rounds accept 6,556 and 6,685 on the two bases. It is not system accuracy. Acceptance rises across self-harvest rounds while the evaluation gain of the second round is 0.4 points and not significant, so the loop is reported at its measured ceiling rather than iterated further.

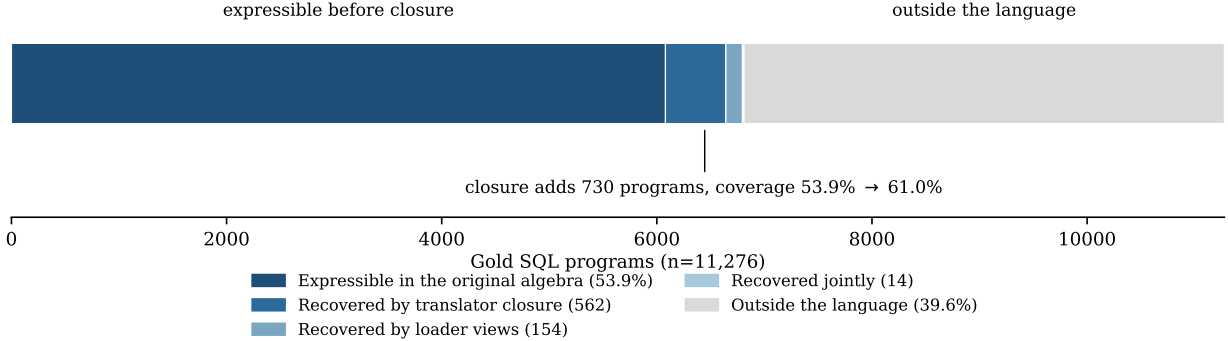


Figure 4: Attribution of WikiTableQuestions program coverage. Compiler and loader closure raise measured coverage from 53.9% to 61.0%, while 39.6% remains outside the language. Of the 730 recovered programs, 562 come from translator closure, 154 from loader views, and 14 require both.

Table 4: Composition of the sealed PACS set.

Gold status	Rows
Answerable, serialised value oracle	655
Answerable, oracle serialisation failed	39
Unsupported	138
No result	90
Total	922

C Candidate Selection

Four candidate programs are sampled per development question after the basic program checks. A perfect selector would gain about five points over choosing the first accepted candidate. The deterministic scorer uses only execution features such as empty results, zero-row filter probes, and answer-kind mismatches. It gains 2.0 points (Figure 6). A faithful LLM selector gains 0.0, and an immediate LLM score loses 0.7. This experiment evaluates candidate ranking and not the main single-call system.

D Reproduction Details

Students are Qwen3-8B (Qwen Team, 2025) and Llama-3.1-8B. Both are adapted with rank-64 LoRA under 4-bit QLoRA quantisation in LLaMA-Factory (Hu et al., 2022; Dettmers et al., 2023; Zheng et al., 2024). Serving uses vLLM with guided JSON decoding compiled from the algebra grammar (Kwon et al., 2023). Candidate collection uses positive-temperature sampling, while evaluation uses one temperature-zero call. Training and evaluation ran on single 80GB A100 and 96GB H100 devices. Each fine-tuning run completes within a few hours on one device, every sealed evaluation is a single pass, the teacher harvest comprises one pass over the 9,267-question training pool, and the final-test teacher replicate adds roughly 3,000 further teacher calls. The WikiTableQuestions official run used a recorded manifest containing the commit hash, file checksums, decoder settings, and adapter identity. An experiment ledger records every run behind this paper, and any rate-style number enters the ledger only after ten deterministically sampled raw transcripts, passes and failures both, have been read by a human. Prompts, manifests, tests, and the experiment ledger are released with the code.

Table 5: Sealed data-access ablation on the final test (n=2,285). Aggregation pools the six exhaustive families, count, sum, comparison, set, top-*k* and min/max (n=1,041); single record pools the remaining answerable types (n=884); refusal pools the three refusal classes (n=360). All values are accuracy in percent.

System	Overall	Answerable	Aggregation	Single record	Refusal
Closed book	16.6	1.0	0.0	2.3	100.0
Retrieval reader	39.2	32.6	3.7	66.5	74.7
with gold constraints	25.6	11.8	3.3	21.9	98.9
Qwen fully local	85.6	83.0	85.7	79.9	99.7

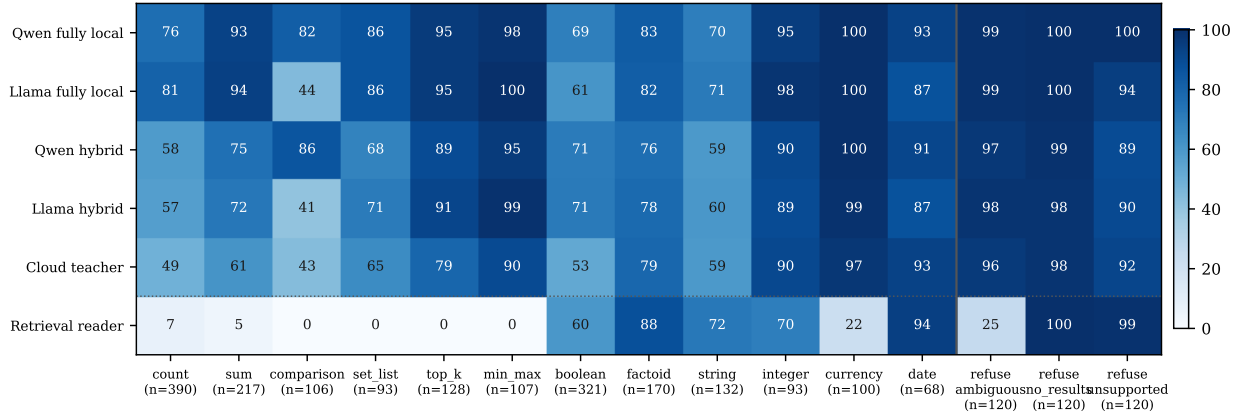


Figure 5: Full sealed final-test accuracy by system and question type, twelve answerable types and three refusal classes, with the sealed retrieval reader of Table 5 as the bottom row.

Table 6: Development ladder and candidate acceptance rates.

Arm	Qwen3-8B	Llama-3.1-8B
Zero-shot scaffolded	70.4	60.0
Supervised fine-tuning	81.2	83.1
Rejection-sampling fine-tuning	81.5	82.7
Preference optimisation	83.5	77.3
Fully local, best arm	86.2	84.2
Teacher candidate acceptance	65.5	
Self-harvest acceptance, round 1	76.6	78.1
Self-harvest acceptance, round 2	86.3	—

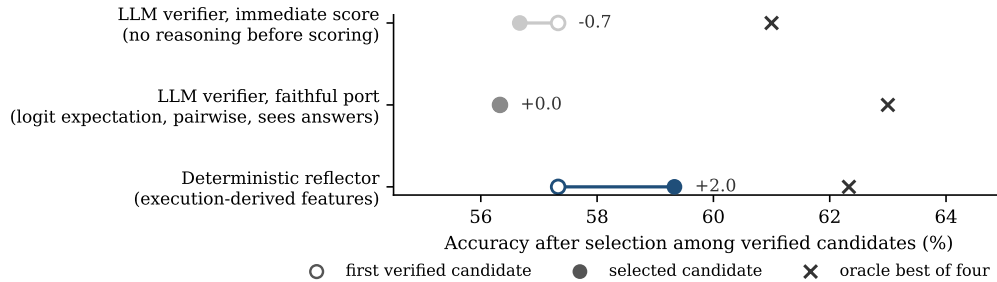


Figure 6: Candidate selection on development data. The deterministic scorer is the only selector that improves over the first accepted candidate. Crosses show the oracle best of four.